

Advanced Software Architecture and Analysis (E26)

Exam and Task

Agenda

- Case
- Report

Introduction

The aim is to study and work with different architectural design solutions, integrating heterogeneous methods, languages, and styles.

The domain we will be focusing on is the Industry 4.0 production domain as an example domain to design a complex system.

You are welcome to choose another application domain but you must be aware of the requirements to the system (will be presented later)

The following example is from the I4.0 domain to illustrate a complex system

Introduction

The Industry 4.0 production domain is characterized by being both physical and including different kinds of information technology, communication channels, and computational devices.

One of the architectural challenges in a production system is to design how production components interact with each other to achieve some common task while ensuring dependability, trustworthiness, and other essential qualities to support a flexible yet reliable production system.

An Industry 4.0 production system contains many different components, e.g.

- Connectors/adapters to production components and machinery
- Coordinators of the components and machinery
- Optimizers of the overall production flow.

Introduction

Typically, a production system already contains different components from different vendors and technologies, which needs to be considered when new components are designed for the system.

It is unavoidable that some technologies are chosen already, which might constrain the architecture in future development.

Also, it is impossible to be up to date with all the technologies existing, and therefore, it raises the need for architects to be acquainted with the characteristics and consequences of design decisions and trade-offs choosing a given technology.

Apply integrated methods and techniques acquired from software architecture with formal reasoning.

#sdiidk



Architectural prototype

The task is to analyze, design and implement architecture prototypes of production system components that interact to achieve some common task as well as to verify essential quality aspects of the system.

An architecture prototype means that it is not necessary to implement all aspects of the system but just enough to be able to show that the architecture can fulfill the stated requirements

There are two types of requirements to the system in the course:

- Requirements to the system
- Requirements from the course

Requirements to the system

Following production system requirements need to be satisfied in the design:

- Production software must be able to exchange and coordinate information to execute a production and change production
- Production software must run 24/7
- Production software must be continuously deployable
- Identify essential quality properties/requirements derived from the examples above.

The design must address a number of quality attributes and behavioral functionalities.

If you choose not to work with an I4.0 system, you must specify similar requirements to the system

Requirements from the course

Part of the design solution must consider, utilize, establish, and implement

- different programming languages
- different databases
- different message buses
- different containers
- Point-to-point communication between two or more programming languages, e.g. protocol buffers
- different architectural styles, e.g. event driven, client-server, microservice, layers
- different architectural patterns/tactics, e.g. circuit breaker, visitor pattern
- requirement representation in architectural description(s) and its formalization
- structural and behavioral models in the system architecture
- formal modeling: semantic anchoring with model transformation
- formal analysis and refinement: verification and validation of the architecture models

Outcome

Research questions the project needs to answer:

1. How can different architectures support the stated system requirements?
2. Which architectural trade offs must be taken due to the technology choices?
3. Which properties can be formally specified, verified and validated, what are the results, and how the architecture can be refined and improved based on the V&V results?

Outcome (handed in as a group)

1. Group report with a maximum length of 10 pages, in a similar format to a scientific paper (pdf and compilable latex code).
2. Zip-file: Github repository with source files (code, model, requirement queries, etc.) and guide how to use (and link to the repository).

Templates

1. Templates for
 1. Report
2. Distributed as Github repository to be cloned (will contain the needed structure and latex files to get started)

Group Report Template

→ The work must be documented in maximum 10 double column pages using the latex template:

- Introduction and motivation
- Problem and approach
- Related work
- Use case and QAS
- Design and analysis modeling
- Formal analysis and refinement including verification and validation (V&V)
- Evaluation (Empirical)
- Conclusion including discussion and future work

Group Report Template

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Abstract—
*Index Terms—*Keyword1, Keyword2, Keyword3, Keyword4, Keyword5

I. INTRODUCTION AND MOTIVATION

The structure of the paper is as follows. Section I outlines the research question and the research approach. Section II describes similar work in the field and how our contribution fits the field. Section III presents a production reconfiguration use case. The use case serves as input to specify a reconfigurability QA requirement in Section IV-B. Section V introduces the proposed reconfigurable middleware software architecture design. Section VI evaluates the proposed middleware on realistic equipment in the I4.0 lab and analyzes the results against the stated QA requirement.

II. PROBLEM AND APPROACH

Problem.

Research questions:

- 1)
- 2)

Approach. The following steps are taken to answer this paper's research questions:

- 1)

III. RELATED WORK

This Section addresses existing contributions by examining xxx in the I4.0 domain. In total, x papers are investigated.

In [1], experiences are elaborated on a three-layer architecture of a reconfigurable smart factory for drug packing in healthcare I4.0.

The paper [2] proposes an ontology agent-based architecture for inferring new configurations to adapt to changes in

IV. USE CASE AND QUALITY ATTRIBUTE SCENARIO

This Section introduces the use case and the specified x QASes. The QASes are developed based on the use case.

A. Use case

B. Quality attribute scenarios

V. THE SOLUTION

This section will describe a proposed design of that aims to achieve the stated QASes stated in the previous section.

VI. EVALUATION

This Section describes the evaluation of the proposed design. Section VI-A introduces the design of the experiment to evaluate the system. Section VI-B identifies the measurements in the system for the experiment. Section VI-C describes the pilot test used to compute the number of replication in the actual evaluation. Section VI-D presents the analysis of the results from the experiment.

A. Experiment design

B. Measurements

C. Pilot test

D. Analysis

VII. FUTURE WORK

VIII. CONCLUSION

REFERENCES

- [1] J. Wan, S. Tang, D. Li, M. Imran, C. Zhang, C. Liu, and Z. Pang, "Reconfigurable smart factory for drug packing in healthcare industry 4.0," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 1, pp. 507–516, 2019.
- [2] Y. Alsafi and V. Vyatkin, "Ontology-based reconfiguration agent for intelligent mechatronic systems in flexible manufacturing," *Robotics and Computer-Integrated Manufacturing*, vol. 26, no. 4, pp. 381–391,

Group Report Structure For the Exam

→ Abstract:

→ Briefly describe introduction to the topic, what is the gap, aim, approach, and results

→ Introduction and motivation:

→ Introduction and motivation to the problem domain.

→ Problem, research questions, and approach:

→ What is the problem to be solved with the reliable architecture you build, and how will the problem be addressed. *The stated problem leads to the stated research question.

→ Related work:

→ The related work should review the state of the art consisting of 8-10 papers and should contextualize how this study provides new knowledge to the field. Here you can combine the work from scientific methods.

→ Use case:

→ Unfold the problem with a use case and describe what the use case is about.

Group Report Structure For the Exam

→ **Quality attribute scenario:**

- The use case is the foundation to describe and specify architectural requirements.

→ **Design and analysis modeling:**

- Describe the design and argue for the design decision and how it meets the QASes. Part of the design decision must specify which tactics/patterns are used (provide arguments) and the trade-offs.
- Analyze the design and illustrate the system architecture model, encompassing both structure and behavior, using formal reasoning.

→ **Formal specification, analysis, and refinement including V&V**

- Formalize the system requirements, including functional properties associated with quality attributes.
- Perform formal V&V and simulation to determine if the system meets the specified requirements. Provide the V&V results, along with any counter-examples, and architecture design refinement based on the results.

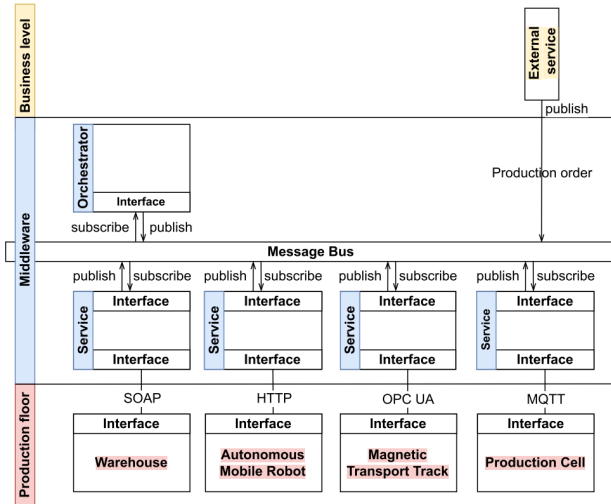
→ **Evaluation:**

- Describe the evaluation design, measurements of the QASes, pilot test, and an analysis of the results. Describe the design for the evaluation, measurements of the QASes, pilot test, and an analysis of the results. From the analysis, how it answers the research questions must be clear.

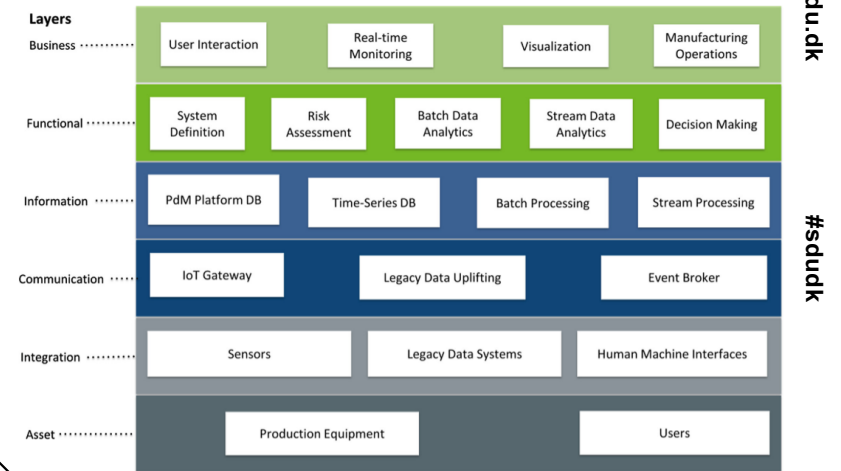
→ **Conclusion: Discussion & Future work:**

- Summary of the work and result.
- Discuss how the work can be extended with respect to the approach and/or evaluation

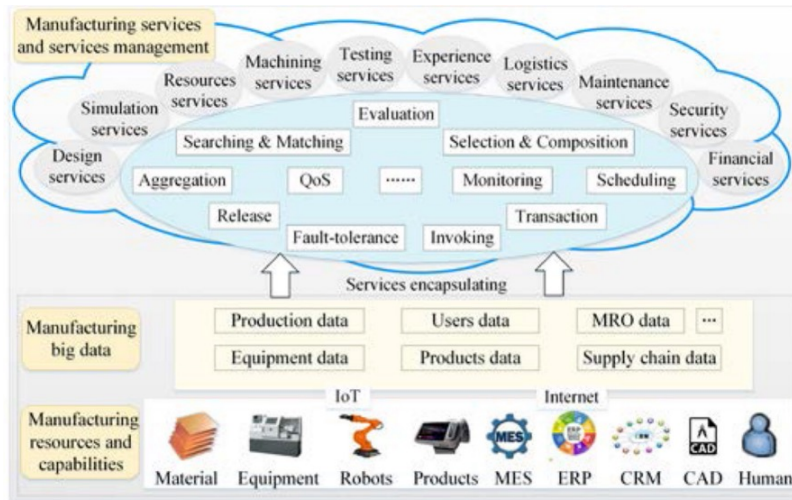
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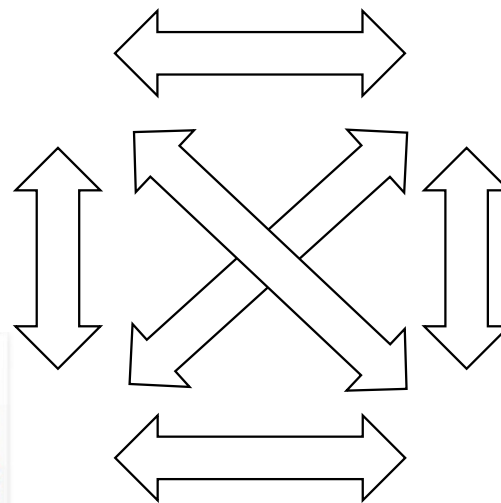
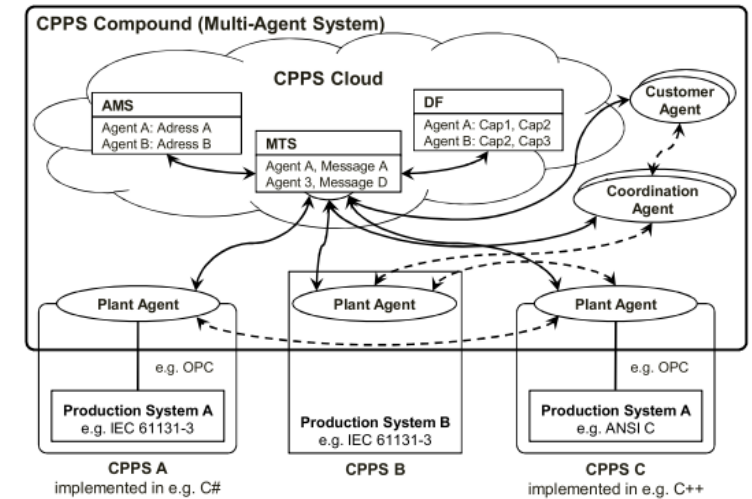
Factory 2



Factory 3



Factory 4



Practicalities

Hardware

→ Use your own PC

Groups

→ 5 members per group

→ You are encouraged to mingle with others

→ Be kind to invite new students to the group

Some friendly advice

- Make good working habits from the beginning, e.g.
 - Commit and build every time you make any changes (think DevOps)
 - Build all you outputs from the beginning
- Use vector graphics in your latex documents
- Be aware that the exam and grading is individual, i.e.
 - Be considerate to each other and don't "devide and conquer"
 - Document who does what

Principles for working with the case

- The basic idea is to be able to **recognize and resolve uncertainties** in problem statements.
- This means that parts of the assignments are **intentionally left incomplete or lacking information**
- Which requires the you to **consider the alternatives and to make a reasonable decision** on the omitted details.
- Why?
 - To enhance learning
 - To encourage group work
 - To encourage discussion
 - To strengthen decision making capabilities

Adapted from [1]

Whats next

- Next lecture we will present the exercises in greater detail
- Github repository
- Exercise description

Literature

- [1] P. Denny, V. Kumar, S. MacNeil, J. Prather, and J. Leinonen, “Probing the unknown: Exploring student interactions with probeable problems at scale in introductory programming,” in Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1, ITiCSE 2025, (New York, NY, USA), pp. 618–624, Association for Computing Machinery, 2025.